

THE BRIDGE (FORCED)

metallic trace map
 $\mathrm{SL}(2, \mathbb{Z})$ on $\kappa = x^2 + y^2 + z^2 - xyz - 2$



mapping class group
of the once-punctured torus



$\mathcal{N} = 2^*$ class- $\mathcal{S}$ S-duality
(Allegretti--Shan)

*same group, same action,
same character variety*

THE WALL (CONFIRMED)

complex volume $\hat{c}(\rho) = i(\mathrm{Vol} + iCS)$
 $\in \mathbb{C}/4\pi^2\mathbb{Z}$ (dimensionless)

the scale lives in $\hbar \leftrightarrow k$
and the squashing radius

figure-eight: $\mathrm{Vol} = 2.0299$, $CS = 0$
the scale-carrier vanishes

A symmetry identity is not a scale. Structure crosses; magnitude does not.